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Optical actuator, camera module, and camera-mounted device

Abstract

An optical actuator that comprises an inside holder that can hold an optical path bending member, an outside holder supporting the inside holder such that the inside holder can pivot around a first axis, and a drive unit making the inside holder pivot. Either end part of the inside holder is rotatably supported on the outside holder. The drive unit has: an ultrasonic motor supported on the outside holder, and an intermediate part supported on the inside holder. The ultrasonic motor has an oscillator that resonates, and the intermediate part has a fan-shaped contact part that contacts the oscillator.

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Background/Summary

TECHNICAL FIELD
(1) The present invention relates to an optical actuator, a camera module, and a camera-mounted device.
BACKGROUND ART
(2) In the related art, a thin camera-mounted device in which a camera module is mounted, such as a smartphone and a digital camera, has been known. The camera module includes: a lens part including one or more lenses; and an image-capturing element that captures a subject image formed by the lens part.
(3) Further, a camera module including a bending optical system, in which light from a subject along a first optical axis is bent in a direction of a second optical axis through a prism, which is an

optical path bending member, provided at a stage prior to a lens part and is guided to the lens part at a stage subsequent to the prism, has also been proposed (for example, Patent Literature (hereinafter, referred to as "PTL") 1).
(4) The camera module disclosed in PTL 1 includes: a shake-correcting device that corrects camera shake generated in a camera; and an auto-focusing device that performs auto-focusing. Such a camera module includes a shake-correcting actuator and an auto-focusing actuator as optical actuators.
(5) Of these actuators, the shake-correcting actuator includes a first actuator and a second actuator that sway the prism about two different axes. Specifically, the first actuator sways the prism about a swaying axis orthogonal to a plane including a first optical axis and a second optical axis. Further, the second actuator sways the prism about a swaying axis that coincides with the second optical axis.
(6) When camera shake is generated in the camera, the shake-correcting actuator sways the prism under the control of a control unit to perform shake correction. Thus, the camera shake generated in the camera is corrected.
CITATION LIST
Patent Literature
(7) PTL 1
(8) Japanese Patent Application Laid-Open No. 2015-092285
SUMMARY OF INVENTION
Technical Problem
(9) For the camera module as described above, an optical actuator having a novel configuration is desired.
(10) An object of the present invention is to provide an optical actuator, a camera module, and a camera-mounted device each having a novel configuration.
Solution to Problem
(11) One aspect of an optical actuator according to the present invention includes: an inner holder capable of holding an optical path bending member; an outer holder configured to support the inner holder in a swayable manner about a first axis; and a drive unit configured to sway the inner holder. The inner holder is rotatably supported through a bearing part with respect to the outer holder at both end portions of the inner holder in a direction parallel to the first axis. The drive unit includes an ultrasonic motor, which is supported by the outer holder, and an interposition part, which is supported by the inner holder. The ultrasonic motor includes a transducer configured to resonate. The interposition part includes a contact part configured to come into contact with the transducer and having a fan shape.
(12) One aspect of a camera module according to the present invention includes: the optical actuator described above; an optical path bending member held by an inner holder of the optical actuator and configured to bend incident light travelling along a first direction in a second direction orthogonal to the first direction; and an image-capturing element disposed on a side in the second direction from the optical actuator.
(13) One aspect of a camera-mounted device according to the present invention includes: the camera module described above; and a control unit that controls the camera module.
Advantageous Effects of Invention
(14) According to the present invention, it is possible to provide an optical actuator, a camera module, and a camera-mounted device each having a novel configuration.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a perspective view of a camera module according to an embodiment of the present invention;
(2) FIG. **2** is an exploded perspective view of an optical path bending module;
(3) FIG. **3** is an exploded perspective view of the optical path bending module;
(4) FIG. **4** is a cross-sectional view of a base main body taken along line X**1**-X**1** in FIG. **1**;
(5) FIG. **5** is a perspective view of an inner holder and members assembled to the inner holder;
(6) FIG. **6** is a perspective view of the inner holder and the members assembled to the inner holder;
(7) FIG. **7** is a perspective view of an outer holder and members assembled to the outer holder;
(8) FIG. **8** is a perspective view of the outer holder and the members assembled to the outer holder;
(9) FIG. **9** is a cross-sectional view of the outer holder;
(10) FIG. **10** is a perspective view of a drive unit;
(11) FIG. **11**A illustrates an example of a camera-mounted device in which the camera module is mounted;
(12) FIG. **11**B illustrates an example of the camera-mounted device in which the camera module is mounted;
(13) FIG. **12**A illustrates an automobile as the camera-mounted device in which an in-vehicle camera module is mounted; and
(14) FIG. **12**B illustrates the automobile as the camera-mounted device in which the in-vehicle camera module is mounted.
DESCRIPTION OF EMBODIMENTS
(15) Hereinafter, an embodiment according to the present invention will be described in detail with reference to the accompanying drawings. Note that, an optical actuator, a camera module, and a camera-mounted device according to an embodiment to be described later are examples of the optical actuator, the camera module, and the camera-mounted device according to the present invention, and the present invention is not limited by the embodiment.
Embodiment
(16) A camera module according to an embodiment of the present invention will be described with reference to FIGS. **1** to **10**. Hereinafter, an outline of camera module C will be described and then the structures of optical path bending module **2**, lens module **10**, and image-capturing element module **11** that are included in camera module C will be described. Note that, the camera actuator, the camera module, and the camera-mounted device according to the present invention may include all configurations to be described later or may not include some of the configurations.
(17) <Camera Module>
(18) Camera module C is mounted in, for example, smartphone M (see FIGS. **11**A and **11**B), a mobile phone, a digital camera, a notebook computer, a tablet terminal, a portable game machine, and a thin camera-mounted device (such as an in-vehicle camera).
(19) Hereinafter, each component that forms camera module C of the present embodiment will be described based on a state in which each component is incorporated in camera module C. Further, in the description of the structure of camera module C of the present embodiment, an orthogonal coordinate system (X, Y, Z) indicated in each drawing is used.
(20) Camera module C is mounted such that the X direction is the left-right direction, the Y direction is the up-down direction, and the Z direction is the front-rear direction as viewed from the photographer, for example, in a case where the camera-mounted device is configured to take a photograph in practice. Light (incident light) from a subject enters mirror MR of optical path bending module **2** from the + side (plus side) in the Z direction as indicated by long dashed short dashed line α (also referred to as a first optical axis) in FIG. **1**. Mirror MR corresponds to an example of the optical path bending member. Note that, the optical path bending member may be, for example, a prism.

(21) The light (emission light) that has entered mirror MR is bent at an optical path bending surface of mirror MR as indicated by long dashed short dashed line β (also referred to as a second optical axis) in FIG. **1** and is guided to lens part **102** of lens module **10** disposed on the + side of mirror MR in the X direction.
(22) Then a subject image formed by lens part **102** is captured by image-capturing element module **11** (see FIG. **1**) disposed on the + side of lens module **10** in the X direction.
(23) In the present embodiment, the direction from the + side in the Z direction to the − side (minus side) in the Z direction corresponds to an example of the first direction. Further, the direction from the − side in the X direction to the + side in the X direction corresponds to an example of the second direction. Nonetheless, the first direction and the second direction are not limited to those in the present embodiment. The first direction and the second direction may be directions orthogonal to each other.
(24) Camera module C of the present embodiment causes mirror swaying device S incorporated in optical path bending module **2** to sway mirror MR.
(25) Further, camera module C of the present embodiment performs auto-focusing by displacing the lens part in the X direction with an AF device (not illustrated) incorporated in lens module **10**. That is, lens module **10** has an auto-focusing function.
(26) Further, lens part **102** of lens module **10** is a so-called zoom lens that can cope with wide-angle photographing (focal length: short) through telephotographing (focal length: long). Camera module C of the present embodiment moves lens part **102** to a position in the X direction in accordance with the wide-angle photographing or the telephotographing with a zoom device (not illustrated) incorporated in lens module **10**. That is, lens module **10** has a zoom function.
(27) (Optical Path Bending Module)
(28) Hereinafter, optical path bending module **2** will be described. Optical path bending module **2** includes base **3**, mirror MR, and mirror swaying device S.
(29) (Base)
(30) Base **3** will be described with reference to FIGS. **1** to **3**. Base **3** corresponds to an example of a fixed-side member. Base **3** includes base main body **31**, and rear-side plate **36**.
(31) (Base Main Body)
(32) Base main body **31** is made of, for example, synthetic resin or nonmagnetic metal. Base main body **31** is a box-shaped member that is partially open. Light from a subject can pass through an opening part (upper-side opening part **316**) of base main body **31** on the + side in the Z direction to penetrate an interior space of the base. Base **3** as described above is configured to accommodate mirror swaying device S to be described later.
(33) Specifically, base main body **31** includes lower-side wall part **311**, left-side wall part **312***a*, right-side wall part **313***b*, front-side wall part **314**, and upper-side wall part **315**. Further, base main body **31** includes upper-side opening part **316**, front-side opening part **317**, rear-side opening part **318**, and lower-side opening part **321**.
(34) (Lower-Side Wall Part)
(35) Lower-side wall part **311** has a rectangular plate shape parallel to the XY plane. Lower-side wall part **311** forms a bottom part of base main body **31**. Note that, hereinafter, the left-right direction means the left-right direction in a case where optical path bending module **2** is viewed from the + side in the X direction with the + side in the Z direction facing upward. Accordingly, the + side in the Y direction corresponds to the left side, and the − side in the Y direction corresponds to the right side. Further, in optical path bending module **2**, the + side in the X direction corresponds to the front side, and the − side in the X direction corresponds to the rear side. In addition, in optical path bending module **2**, the + side in the Z direction corresponds to the upper side, and the − side in the Z direction corresponds to the lower side.
(36) (Left-Side Wall Part)
(37) Left-side wall part **312***a* has a plate shape that is a rectangular plate shape parallel to the XZ

plane. Left-side wall part **312**_a_ extends to the + side in the Z direction (the upper side) from a left end portion of lower-side wall part **311**. The left-side wall part **312**_a_ includes a lower end portion connected to the left end portion of lower-side wall part **311**.

(38) (Right-Side Wall Part)

(39) Right-side wall part **313**_b_ has a plate shape that is a rectangular plate shape parallel to the XZ plane. Right-side wall part **313**_b_ extends to the + side in the Z direction (the upper side) from a right end portion of lower-side wall part **311**. Right-side wall part **313**_b_ includes a lower end portion connected to the right end portion of lower-side wall part **311**.

(40) (Front-Side Wall Part)

(41) Front-side wall part **314** has a rectangular plate shape parallel to the YZ plane. Front-side wall part **314** includes a left end portion connected to a front end portion of left-side wall part **312**_a_. Front-side wall part **314** includes a right end portion connected to a front end portion of right-side wall part **313**_b_. Further, front-side wall part **314** includes a lower end portion connected to a front end portion of lower-side wall part **311**. Front-side wall part **314** includes an upper end portion connected to a front end portion of upper-side wall part **315** to be described later.

(42) As illustrated in FIG. **4**, left-side power supply terminal part **92** and right-side power supply terminal part **93** in power supply part **9** to be described later are fixed to a rear-side surface of front-side wall part **314**.

(43) (Upper-Side Wall Part)

(44) Upper-side wall part **315** has a rectangular plate shape parallel to the XY plane. Upper-side wall part **315** is provided on an upper side of lower-side wall part **311** and faces lower-side wall part **311** with a predetermined gap therebetween in the up-down direction (the Z direction).

(45) Upper-side wall part **315** includes a left end portion connected to an upper end portion of left-side wall part **312**_a_. Upper-side wall part **315** includes a right end portion connected to an upper end portion of right-side wall part **313**_b_. Upper-side wall part **315** includes a front end portion connected to an upper end portion of front-side wall part **314**.

(46) (Upper-Side Opening Part)

(47) Upper-side opening part **316** is provided at a position including a center portion of upper-side wall part **315**. Upper-side opening part **316** is formed of a through-hole having a substantially rectangular shape which penetrates upper-side wall part **315** in the up-down direction (the Z direction). Incident light passes through upper-side opening part **316** and enters optical path bending module **2**.

(48) (Front-Side Opening Part)

(49) Front-side opening part **317** is provided at a position including a center portion of front-side wall part **314**. Front-side opening part **317** is formed of a through-hole having a rectangular shape which penetrates front-side wall part **314** in the X direction (the front-rear direction). Incident light bent by mirror MR passes through front-side opening part **317** and enters lens module **10**.

(50) (Rear-Side Opening Part)

(51) Rear-side opening part **318** is provided in a rear end portion of base main body **31**. Specifically, rear-side opening part **318** is an opening part having a rectangular shape defined by a rear end portion of left-side wall part **312**_a_, a rear end portion of right-side wall part **313**_b_, a rear end portion of upper-side wall part **315**, and a rear end portion of power supply part **9** (specifically, FPC **90**) to be described later. Rear-side opening part **318** is covered by rear-side plate **36**.

(52) (Rear-Side Plate)

(53) Rear-side plate **36** has a plate shape parallel to the YZ plane. Rear-side plate **36** covers rear-side opening part **318** from the rear. Rear-side plate **36** is fixed to the rear end portion of left-side wall part **312**_a_, the rear end portion of right-side wall part **313**_b_, and the rear end portion of upper-side wall part **315** by a fixing means such as adhesion.

(54) <Mirror Swaying Device>

(55) Next, mirror swaying device S will be described. Mirror swaying device S sways mirror MR

about first axis A**1** (see FIG. **3**) parallel to the Y direction. Mirror swaying device S as such is disposed in an accommodation space included in base **3**.
(56) Mirror swaying device S includes inner holder **4**, outer holder **5**, sway support part **63**, drive unit **8**, and power supply part **9**.
(57) In mirror swaying device S, inner holder **4** is configured to support mirror MR. Inner holder **4** is supported by base **3** through outer holder **5**.
(58) Outer holder **5** supports inner holder **4** through sway support part **63**. Inner holder **4** is swayable about first axis A**1** with respect to outer holder **5**.
(59) When drive unit **8** is driven under the control of control unit **13** (see FIG. **1**), inner holder **4** sways about first axis A**1** with respect to outer holder **5**. Since outer holder **5** is supported by base **3** by means of a support mechanism (not illustrated), inner holder **4** sways about first axis A**1** with respect to base **3**. Then, mirror MR supported by inner holder **4** also sways about first axis A**1** with respect to base **3**. Hereinafter, specific structures of the respective members included in mirror swaying device S will be described.
(60) (Inner Holder)
(61) Inner holder **4** will be described with reference to FIGS. **2**, **3**, **5** and **6**. Inner holder **4** corresponds to an example of a movable-side member and holds mirror MR.
(62) Inner holder **4** includes mount part **41**, left-side plate part **41***a*, right-side plate part **42***b*, and rear-side plate part **43**. Further, inner holder **4** includes left-side shaft part **44***a*, right-side shaft part **45***b*, inner fixing part **46**, and magnet disposing part **47**. Inner holder **4** has a symmetrical shape.
(63) (Mount Part)
(64) Mount part **41** has a rectangular plate shape. Mount part **41** is inclined so as to be located downward (the − side in the Z direction) from a rear end portion thereof (the end portion thereof on the − side in the X direction) towards a front end portion thereof (the end portion thereof on the + side in the X direction). Accordingly, the front end portion of mount part **41** is located downward from the rear end portion of mount part **41**. An upper surface of mount part **41** is a mounting surface configured to dispose mirror MR. In the case of the present embodiment, the inclination direction of mount part **41** is parallel to a direction of a tangent to a circle around first axis A**1**.
(65) (Left-Side Plate Part)
(66) Left-side plate part **41***a* has a plate shape parallel to the XZ plane. Left-side plate part **41***a* extends upward from a left end portion of mount part **41**.
(67) (Right-Side Plate Part)
(68) Right-side plate part **42***b* has a plate shape parallel to the XZ plane. Right-side plate part **42***b* extends upward from a right end portion of mount part **41**.
(69) (Rear-Side Plate Part)
(70) Rear-side plate part **43** has a plate shape parallel to the YZ plane in a state in which inner holder **4** does not sway. Rear-side plate part **43** includes a left end portion connected to a rear end portion of left-side plate part **41***a* (the end portion thereof on the − side in the X direction). Rear-side plate part **43** includes a right end portion connected to a rear end portion of right-side plate part **42***b*. Rear-side plate part **43** includes a lower end portion connected to the rear end portion of mount part **41**.
(71) (Left-Side Shaft Part)
(72) Left-side shaft part **44***a* is configured to support an inner ring of left-side bearing **631***a* in sway support part **63** to be described later. Left-side shaft part **44***a* is provided in an outer surface of left-side plate part **41***a* (the side surface thereof on the + side in the Y direction). Left-side shaft part **44***a* extends from the outer surface of left-side plate part **41***a* to the left side (the + side in the Y direction).
(73) (Right-Side Shaft Part)
(74) Right-side shaft part **45***b* is configured to support an inner ring of right-side bearing **632***b* in sway support part **63** to be described later. Right-side shaft part **45***b* is provided in an outer surface

of right-side plate part **42**_b_ (the side surface thereof on the − side in the Y direction). Right-side shaft part **45**_b_ extends from the outer surface of right-side plate part **42**_b_ to the right side (the − side in the Y direction). The center axis of left-side shaft part **44**_a_ and the center axis of right-side shaft part **45**_b_ are located on the same straight line. The center axis of left-side shaft part **44**_a_ and the center axis of right-side shaft part **45**_b_ coincide with first axis A**1**.
(75) In the case of the present embodiment, a configuration in which left-side bearing **631**_a_ and right-side bearing **632**_b_ in sway support part **63** to be described later are supported by left-side shaft part **44**_a_ and right-side shaft part **45**_b_ is employed, and thus, inner holder **4** can sway about first axis A**1** with high accuracy and stably.
(76) (Inner Fixing Part)
(77) Inner fixing part **46** is a portion to which interposition part **80** of drive unit **8** to be described later is fixed. Inner fixing part **46** is provided in a lower surface of mount part **41**. Inner fixing part **46** is provided at a position, which allows interposition part **80** to be fixed, in a center portion of the lower surface of mount part **41** in the Y direction.
(78) (Magnet Disposing Part)
(79) Magnet disposing part **47** is configured to dispose magnet **83** of drive unit **8** to be described later therein. Magnet disposing part **47** is provided in a left end portion of a rear-side surface of rear-side plate part **43**. Magnet disposing part **47** is formed of a recessed portion of which the + side in the Y direction is open. By employing the configuration of magnet disposing part **47** as such, inner holder **4** can securely hold magnet **83** of drive unit **8**. As a result, it is possible to suppress magnet **83** of drive unit **8** falling off inner holder **4** in a case where inner holder **4** sways.
(80) Note that, in the case of the present embodiment, a simulated magnet disposing part configured in the same manner as magnet disposing part **47** is provided in a right end portion of the rear-side surface of rear-side plate part **43**. This simulated magnet disposing part is configured to cause inner holder **4** to have a symmetrical shape. No magnet is disposed in the simulated magnet disposing part. Nonetheless, a simulated magnet having the same weight as magnet **83** may be disposed in the simulated magnet disposing part. When a configuration in which such a simulated magnet is disposed is employed, the weight balance of inner holder **4** in the left-right direction becomes better.
(81) (Outer Holder)
(82) Outer holder **5** will be described with reference to FIGS. **2**, **3** and **7** to **9**. Outer holder **5** corresponds to an example of the fixed-side member and supports inner holder **4** in a swayable manner about first axis A**1** through sway support part **63** to be described later.
(83) Outer holder **5** has a box shape opening upward (on the + side in the Z direction). Outer holder **5** includes accommodation space **5**_c_ configured to accommodate inner holder **4**.
(84) In the case of the present embodiment, outer holder **5** is supported by base **3** by means of the support mechanism (not illustrated) in a state in which outer holder **5** cannot be displaced with respect to base **3**. Nonetheless, outer holder **5** may be supported by base **3** by means of the support mechanism (not illustrated) in a state in which outer holder **5** can be displaced with respect to base **3**. In this case, outer holder **5** may be supported by base **3** by means of the support mechanism in a state in which outer holder **5** can sway about an axis parallel to the X-axis or the Z-axis. In a case where outer holder **5** is displaced with respect to base **3**, outer holder **5** corresponds to an example of the movable-side member.
(85) Outer holder **5** includes inclined plate part **501**, left-side plate part **502**_a_, right-side plate part **503**_b_, front-side plate part **504**, rear-side plate part **505**, and lower-side plate part **506**. Further, outer holder **5** includes front-side opening part **507**, left-side bearing holding part **508**_a_, right-side bearing holding part **509**_b_, fixing part **514**, and element disposing part **515**.
(86) (Inclined Plate Part)
(87) Inclined plate part **501** has a plate shape and covers accommodation space **5**_c_ from below. Inclined plate part **501** is inclined so as to be located downward (the − side in the Z direction) from

a rear end portion thereof to a front end portion thereof. Accordingly, the front end portion of inclined plate part **501** is located downward from the rear end portion of the inclined plate part **501**.
(88) Inclined plate part **501** as such is provided downward from mount part **41** of inner holder **4**. The inclination angle of inclined plate part **501** with respect to the X direction is the same as the inclination angle of mount part **41** with respect to the X direction. In other words, an upper surface of inclined plate part **501** is parallel to the lower surface of mount part **41**. The upper surface of inclined plate part **501** faces the lower surface of mount part **41** with a predetermined gap therebetween in the up-down direction. A space configured to dispose drive unit **8** to be described later is provided between the upper surface of inclined plate part **501** and the lower surface of mount part **41**.
(89) (Left-Side Plate Part)
(90) Left-side plate part **502***a* has a plate shape parallel to the XZ plane and covers accommodation space **5***c* from the left side (the + side in the Y direction). A portion of left-side plate part **502***a* is connected to a left end portion of inclined plate part **501**. Left-side plate part **502***a* is located on the left side of left-side plate part **41***a* of inner holder **4** (on the + side thereof in the Y direction). Left-side plate part **502***a* faces left-side plate part **41***a* of inner holder **4** in the left-right direction (the Y direction).
(91) (Right-Side Plate Part)
(92) Right-side plate part **503***b* has a plate shape parallel to the XZ plane and covers accommodation space **5***c* from the right side (the − side in the Y direction). A portion of right-side plate part **503***b* is connected to a right end portion of inclined plate part **501**. Right-side plate part **503***b* faces left-side plate part **502***a* with a predetermined gap therebetween in the Y direction. Right-side plate part **503***b* is located on the right side of right-side plate part **42***b* of inner holder **4** (on the − side thereof in the Y direction). Right-side plate part **503***b* faces right-side plate part **42***b* of inner holder **4** in the left-right direction (the Y direction).
(93) (Front-Side Plate Part)
(94) Front-side plate part **504** has a plate shape parallel to the YZ plane and covers accommodation space **5***c* from the front side (the + side in the X direction). Front-side plate part **504** includes a left end portion connected to a front end portion of left-side plate part **502***a*. Front-side plate part **504** includes a right end portion connected to a front end portion of right-side plate part **503***b*. Front-side plate part **504** includes a front-side surface that faces the rear-side surface of front-side wall part **314** in base main body **31** with a predetermined gap therebetween in the X direction.
(95) (Rear-Side Plate Part)
(96) Rear-side plate part **505** has a plate shape parallel to the YZ plane and covers accommodation space **5***c* from the rear side (the − side in the X direction). Rear-side plate part **505** includes a left end portion connected to a rear end portion of left-side plate part **502***a* (the end portion thereof on the − side in the X direction). Rear-side plate part **505** includes a right end portion connected to a rear end portion of right-side plate part **503***b*. Rear-side plate part **505** is connected to a rear end portion (upper end portion) of inclined plate part **501**.
(97) (Lower-Side Plate Part)
(98) Lower-side plate part **506** is a plate-like member and connects the front end portion (lower end portion) of inclined plate part **501** and a lower end portion of front-side plate part **504** in the X direction. Lower-side plate part **506** includes a rear end portion connected to the front end portion (lower end portion) of inclined plate part **501**. Lower-side plate part **506** includes a front end portion connected to the lower end portion of front-side plate part **504**.
(99) (Front-Side Opening Part)
(100) Front-side opening part **507** is provided at a position including a center portion of front-side plate part **504**. Front-side opening part **507** is formed of a through-hole having a rectangular shape which penetrates front-side plate part **504** in the X direction (the front-rear direction). Front-side opening part **507** is provided so as to overlap front-side opening part **317** of base main body **31** in

the X direction. Incident light bent by mirror MR passes through front-side opening part **507** and front-side opening part **317** and enters lens module **10**.
(101) (Left-Side Bearing Holding Part)
(102) Left-side bearing holding part **508***a* is configured to hold left-side bearing **631***a* of sway support part **63** to be described later. Left-side bearing holding part **508***a* is provided in an inner surface of left-side plate part **502***a* (the side surface thereof on the − side in the Y direction). Specifically, left-side bearing holding part **508***a* is formed of a recessed portion provided in the inner surface of left-side plate part **502***a*. An outer ring of left-side bearing **631***a* is fixed to left-side bearing holding part **508***a* by a fixing means such as adhesion.
(103) (Right-Side Bearing Holding Part)
(104) Right-side bearing holding part **509***b* is configured to hold right-side bearing **632***b* of sway support part **63** to be described later. Right-side bearing holding part **509***b* is provided in an inner surface of right-side plate part **503***b* (the side surface thereof on the + side in the Y direction). Specifically, right-side bearing holding part **509***b* is formed of a recessed portion provided in the inner surface of right-side plate part **503***b*.
(105) Right-side bearing holding part **509***b* faces left-side bearing holding part **508***a* in the Y direction. An outer ring of right-side bearing **632***b* is fixed to right-side bearing holding part **509***b* as such by a fixing means such as adhesion.
(106) (Fixing Part)
(107) Fixing part **514** is a portion to which ultrasonic motor **81** of drive unit **8** to be described later is fixed. Fixing part **514** is provided in a rear end portion of the upper surface of inclined plate part **501** (the end portion thereof on the − side in the X direction).
(108) (Element Disposing Part)
(109) Element disposing part **515** is a portion configured to dispose position detection element **82** of drive unit **8** therein. Element disposing part **515** is formed of a through-hole provided in left-side plate part **502***a*. This through-hole penetrates left-side plate part **502***a* in the Y direction. Element disposing part **515** faces magnet disposing part **47** of inner holder **4** in the Y direction.
(110) (Sway Support Part)
(111) Sway support part **63** corresponds to an example of the bearing part and is configured to support inner holder **4** with respect to outer holder **5** in a state in which inner holder **4** is swayable about first axis A**1**. Sway support part **63** includes left-side bearing **631***a* and right-side bearing **632***b*.
(112) (Left-Side Bearing)
(113) Left-side bearing **631***a* is configured to rotatably support a left end portion of inner holder **4** (the end portion thereof on the + side in the Y direction) with respect to outer holder **5**. Left-side bearing **631***a* is a so-called radial bearing. The center axis of left-side bearing **631***a* coincides with first axis A**1**.
(114) Left-side bearing **631***a* includes: an inner ring including an inner ring raceway in an outer peripheral surface of the inner ring; an outer ring including an outer ring raceway in an inner peripheral surface of the outer ring; and a plurality of rolling elements (for example, balls) rollably provided between the inner ring raceway and the outer ring raceway. Each rolling element may be held by a holder. The inner ring of left-side bearing **631***a* is fixed to left-side shaft part **44***a* of inner holder **4** by, for example, tight fit. The outer ring of left-side bearing **631***a* is fixed to left-side bearing holding part **508***a* of outer holder **5**.
(115) (Right-Side Bearing)
(116) Right-side bearing **632***b* is configured to rotatably support a right end portion of inner holder **4** (the end portion thereof on the − side in the Y direction) with respect to outer holder **5**. Right-side bearing **632***b* is a so-called radial bearing. The center axis of right-side bearing **632***b* coincides with first axis A**1**.
(117) Right-side bearing **632***b* includes: an inner ring including an inner ring raceway in an outer

peripheral surface of the inner ring; an outer ring including an outer ring raceway in an inner peripheral surface of the outer ring; and a plurality of rolling elements (for example, balls) rollably provided between the inner ring raceway and the outer ring raceway. Each rolling element may be held by a holder. The inner ring of right-side bearing **632***b* is fixed to right-side shaft part **45***b* of inner holder **4** by, for example, tight fit. The outer ring of right-side bearing **632***b* is fixed to right-side bearing holding part **509***b* of outer holder **5**.
(118) In the case of the present embodiment, inner holder **4** is supported by the radial bearings with respect to outer holder **5**, and thus, it is possible to suppress inclination of the sway center of inner holder **4** with respect to first axis A**1**. As a result, inner holder **4** can sway with high accuracy.
(119) (Drive Unit)
(120) Drive unit **8** is a drive unit configured to sway inner holder **4** about first axis A**1** with respect to outer holder **5**. Drive unit **8** is provided between inner holder **4** and outer holder **5**. Drive unit **8** is provided so as to be along the lower surface of mount part **41** in inner holder **4** and the upper surface of inclined plate part **501** in outer holder **5**. Such a disposition aspect of drive unit **8** contributes to a reduction in the size of optical path bending module **2**.
(121) Drive unit **8** as such includes interposition part **80**, ultrasonic motor **81**, position detection element **82** (see FIG. **8**), and magnet **83** (see FIG. **5**) as illustrated in FIG. **10**.
(122) (Interposition Part)
(123) Interposition part **80** includes holder **801** and a pair of contact parts **802** and **803**. Holder **801** is a member which is made of, for example, metal, ceramic, or synthetic resin and to which the pair of contact parts **802** and **803** is fixed. In the case of the present embodiment, holder **801** is a cuboid. Nonetheless, the shape of holder **801** is not limited to that in the case of the present embodiment. Holder **801** is fixed to inner fixing part **46** of inner holder **4** by a fixing means such as adhesion.
(124) Each of the pair of contact parts **802** and **803** has, for example, a plate shape made of metal. In the case of the present embodiment, contact part **802** of the pair of contact parts **802** and **803** has, as a shape thereof viewed in the Y direction, a fan shape partially including arc part **802***a*, and contact part **803** of the pair of contact parts **802** and **803** has, as a shape thereof viewed in the Y direction, a fan shape partially including arc part **803***a*.
(125) Note that, in the present embodiment, the fan shape means at least a shape with an arc-shaped appearance centered on a predetermined center point. In the case of the present embodiment, the predetermined center point is present on first axis A**1**. That is, arc parts **802***a* and **803***a* are each formed of an arc centered on the predetermined center point present on first axis A**1**. The predetermined center point, however, may be deviated from first axis A**1**.
(126) When inner holder **4** sways about first axis A**1**, each of arc parts **802***a* and **803***a* of the pair of contact parts **802** and **803** moves along an arc around first axis A**1**. Such a configuration makes it possible to stably bring the pair of contact parts **802** and **803** into contact with transducers **812** and **813** of resonance part **810** in the entire sway stroke range of inner holder **4** while reducing the size of the pair of contact parts **802** and **803**.
(127) The pair of contact parts **802** and **803** is fixed to holder **801** at each upper end portion of the pair of contact parts **802** and **803** (each end portion thereof on the + side in the Z direction). Contact parts **802** and **803** are provided adjacent to each other in the Y direction. Contact part **802** is provided on the left side of contact part **803** (on the + side thereof in the Y direction).
(128) The pair of contact parts **802** and **803** as such is portions with which transducers **812** and **813** of ultrasonic motor **81** to be described later come into contact, respectively. Specifically, transducer **812** is in contact with a left-side surface of contact part **802** (the side surface thereof on the + side in the Y direction), and transducer **813** is in contact with a right-side surface of contact part **803** (the side surface thereof on the − side in the Y direction).
(129) When interposition part **80** is configured as described above, thrust in a direction of a tangent to a circle around first axis A**1** is applied from ultrasonic motor **81** (specifically, transducers **812**

and **813**) to interposition part **80** (specifically, the pair of contact parts **802** and **803**). Then, inner holder **4** sways about first axis A**1** based on the thrust.
(130) Note that, in the case of the present embodiment, interposition part **80** includes the pair of contact parts **802** and **803**. Nonetheless, the number of contact parts included in interposition part **80** is not particularly limited. For example, interposition part **80** may include only one contact part of the pair of contact parts **802** and **803** (that is, a contact part). In this case, the position of the contact part may be adjusted as appropriate.
(131) (Ultrasonic Motor)
(132) Ultrasonic motor **81** is a driving source that generates a driving force for swaying inner holder **4** about first axis A**1**. Ultrasonic motor **81** is fixed to fixing part **514** in outer holder **5**. Ultrasonic motor **81** is disposed so as to be along the upper surface of inclined plate part **501**.
(133) Hereinafter, in the description of ultrasonic motor **81**, a direction of an intersection line between the upper surface of inclined plate part **501** and a plane parallel to the XZ plane will be referred to as the U direction (see FIG. **9**). The + side in the U direction is a direction from the rear end portion of inclined plate part **501** (the end portion thereof on the − side in the X direction) to the front end portion thereof (the end portion thereof on the + side in the X direction). The − side in the U direction is a direction from the front end portion of inclined plate part **501** (the end portion thereof on the + side in the X direction) to the rear end portion thereof (the end portion thereof on the − side in the X direction). Note that, the U direction is also a direction of a tangent to a circle around first axis A**1**.
(134) Further, a direction of a normal to the upper surface of inclined plate part **501** will be referred to as the V direction (see FIG. **9**). The + side in the V direction is a direction from a lower-side surface of inclined plate part **501** (the side surface thereof on the − side in the Z direction) to an upper-side surface thereof (the side surface thereof on the + side in the Z direction). The − side in the V direction is a direction from the upper-side surface of inclined plate part **501** (the side surface thereof on the + side in the Z direction) to the rear end portion thereof (the side surface thereof on the + side in the Z direction).
(135) Ultrasonic motor **81** includes resonance part **810**, a pair of piezoelectric elements **815** and **816**, and first electrode **817**.
(136) (Resonance Part)
(137) Resonance part **810** is formed of, for example, a conductive material and resonates with vibration of the pair of piezoelectric elements **815** and **816** to convert vibrational motion into linear motion in a predetermined direction (the U direction in the present embodiment). Resonance part **810** is disposed so as to be along the U direction (in other words, a tangent to a circle around first axis A**1**).
(138) When resonance part **810** vibrates based on the vibration of the pair of piezoelectric elements **815** and **816**, a driving force in the U direction (the + side in the U direction or the − side in the U direction) acts on interposition part **80** (specifically, the pair of contact parts **802** and **803**) from resonance part **810**. Interposition part **80** then sways about first axis A**1**. As a result, inner holder **4** to which interposition part **80** is fixed sways about first axis A**1**.
(139) Resonance part **810** has at least two resonance frequencies (a first resonance frequency and a second resonance frequency) and is deformed with different behaviors with respect to the resonance frequencies, respectively. In other words, the entire shape of resonance part **810** is set such that resonance part **810** is deformed with different behaviors with respect to the two resonance frequencies.
(140) The different behaviors are a behavior that gives a force towards one direction in the U direction (for example, the + side in the U direction) with respect to interposition part **80** and a behavior that gives a force towards the other direction in the U direction (for example, the − side in the U direction) with respect to interposition part **80**. In other words, the different behaviors are a behavior that sways inner holder **4** in the first direction about first axis A**1** and a behavior that

sways inner holder **4** in the second direction about first axis A**1**.
(141) Resonance part **810** as such includes trunk part **811**, a pair of transducers **812** and **813**, and energization part **814**.
(142) Trunk part **811** has a substantially rectangular plate shape.
(143) The pair of transducers **812** and **813** corresponds to examples of a first transducer and a second transducer and extends forward (to the + side in the U direction) from a front end portion of trunk part **811**. In the case of the present embodiment, each of the pair of transducers **812** and **813** has a plate shape and is formed integrally with trunk part **811**.
(144) The pair of transducers **812** and **813** has a symmetrical shape with respect to a plane parallel to the UV plane (the XZ plane). Transducers **812** and **813** are adjacent to each other with a predetermined gap therebetween in the Y direction. Transducer **812** is provided on the left side of transducer **813** (on the + side thereof in the Y direction).
(145) The pair of transducers **812** and **813** each includes a base end portion connected to the front end portion of trunk part **811** (the end portion thereof on the + side in the U direction). The pair of transducers **812** and **813** each includes a leading end portion as a free end. The leading end portion of transducer **812** is in contact with the left-side surface of contact part **802**. The leading end portion of transducer **813** in in contact with the right-side surface of contact part **803**. Note that, in case where interposition part **80** includes only one contact part of the pair of contact parts **802** and **803** (that is, a contact part), the leading end portion of transducer **812** may be in contact with a left-side surface of the one contact part and the leading end portion of transducer **813** may be in contact with a right-side surface of the one contact part.
(146) In the case of the present embodiment, resonance part **810** includes the pair of transducers **812** and **813**. Nonetheless, the number of transducers included in resonance part **810** is not particularly limited. For example, resonance part **810** may include only one transducer of the pair of transducers **812** and **813** (that is, a transducer). In this case, the position of the transducer may be adjusted as appropriate.
(147) Energization part **814** extends from a rear end portion of trunk part **811** (the end portion thereof on the − side in the U direction) to the − side in the U direction. Energization part **814** is fixed to fixing part **514** in outer holder **5** by a fixing means such as adhesion. Further, energization part **814** is connected to a predetermined terminal of right-side power supply terminal part **93** of power supply part **9** to be described later through a right-side connection part (not illustrated). Among end portions of a power supply path thereof, an end portion connected to energization part **814** forms a second electrode (not illustrated) of ultrasonic motor **81**.
(148) (Piezoelectric Element)
(149) Each the pair of piezoelectric elements **815** and **816** is, for example, a plate-shaped vibration element made of ceramic. Each of the pair of piezoelectric elements **815** and **816** vibrates when a high-frequency voltage is applied thereto. The pair of piezoelectric elements **815** and **816** is provided so as to be adjacent to each other in the V direction. Piezoelectric element **815** is located upward (on the + side in the V direction) from piezoelectric element **816**.
(150) The pair of piezoelectric elements **815** and **816** sandwiches trunk part **811** of resonance part **810** in the V direction. Accordingly, the pair of piezoelectric elements **815** and **816** and resonance part **810** (trunk part **811**) are electrically connected to each other.
(151) (First Electrode)
(152) First electrode **817** includes sandwiching part **818** that sandwiches the pair of piezoelectric elements **815** and **816** in the V direction. The pair of piezoelectric elements **815** and **816** and first electrode **817** (sandwiching part **818**) are electrically connected to each other. Accordingly, the pair of piezoelectric elements **815** and **816**, resonance part **810**, and first electrode **817** are electrically connected to each other.
(153) Further, first electrode **817** includes electrode part **819** to which a voltage is applied. Electrode part **819** is electrically connected to a terminal of right-side power supply terminal part

93 of power supply part **9** to be described later through the right-side connection part (not illustrated). When a voltage is applied to electrode part **819**, first electrode **817** as such applies a voltage to the pair of piezoelectric elements **815** and **816** through sandwiching part **818**.
(154) (Position Detection Element and Magnet)
(155) Position detection element **82** is disposed in element disposing part **515** of outer holder **5**. Position detection element **82** is connected to a power supply terminal that forms power supply part **9** to be described later.
(156) Further, magnet **83** is a cuboid. Magnet **83** as such is magnetized in the Z direction. Magnet **83** is fixed to magnet disposing part **47** of inner holder **4**.
(157) Position detection element **82** is provided on the + side of magnet **83** in the Y direction. Position detection element **82** faces magnet **83** with a predetermined gap therebetween in the Y direction. Position detection element **82** detects the magnetic flux (also referred to as position information) of magnet **83** and transmits a detection value to control unit **13** implemented in sensor board **12**.
(158) Specifically, in the case of the present embodiment, position detection element **82** detects a change in the magnetic flux passing through a detection surface of position detection element **82** in the Y direction. The dispositions of magnet **83** and position detection element **82** are set such that the magnetic flux passing through position detection element **82** in the Y direction becomes zero in a state in which inner holder **4** does not sway (this state will also be referred to as the reference state of the inner holder). When inner holder **4** sways about first axis A**1** from the reference state, the magnetic flux passing through position detection element **82** in the X direction changes.
(159) Control unit **13** (see FIG. **1**) determines the location of magnet **83** (that is, inner holder **4**) around first axis A**1** based on the detection value received from position detection element **82**.
(160) (Regarding Operation of Drive Unit)
(161) In the case of drive unit **8** having the configuration as described above, the pair of piezoelectric elements **815** and **816** vibrates when a voltage is applied to ultrasonic motor **81** through power supply part **9** to be described later under the control of control unit **13**. In a case where the frequency of the vibration generated by the pair of piezoelectric elements **815** and **816** corresponds to the first resonance frequency set to resonance part **810**, resonance part **810** gives a force towards one direction in the U direction (for example, the + side in the U direction) to interposition part **80**. As a result, inner holder **4** sways in the first direction about first axis A**1** with respect to outer holder **5**.
(162) In a case where the frequency of the vibration generated by the pair of piezoelectric elements **815** and **816** corresponds to the second resonance frequency set to resonance part **810**, on the other hand, resonance part **810** gives a force towards the other direction in the U direction (for example, the − side in the U direction) to interposition part **80**. As a result, inner holder **4** sways in the second direction about first axis A**1** with respect to outer holder **5**.
(163) (Power Supply Part)
(164) Power supply part **9** will be described with reference to FIGS. **2** to **4**. Power supply part **9** is configured to supply power (apply a voltage) to drive unit **8**. Specifically, power supply part **9** supplies power to position detection element **82** of drive unit **8** and to ultrasonic motor **81** of drive unit **8**.
(165) Power supply part **9** as such includes FPC **90**, left-side power supply terminal part **92**, right-side power supply terminal part **93**, a left-side connection part (not illustrated), and the right-side connection part (not illustrated).
(166) (FPC)
(167) FPC **90** is a flexible printed circuit board and is connected to a power supply (not illustrated) of a camera-mounted device (hereinafter, this power supply will be simply referred to as “power supply”). FPC **90** has a substantially rectangular plate shape.
(168) FPC **90** as such includes outer terminal part **901**, inner first terminal part **902**, and inner

second terminal part **903**.
(169) (Outer Terminal Part)
(170) Outer terminal part **901** is provided in a left end portion of FPC **90**. In the case of the present embodiment, outer terminal part **901** includes a plurality of terminals. These respective terminals are connected to the power supply and are also connected to ultrasonic motor **81** and position detection element **82**.
(171) (Inner First Terminal Part)
(172) Inner first terminal part **902** is provided in a left end portion of a front end portion of FPC **90**. In the case of the present embodiment, inner first terminal part **902** includes a plurality of terminals. These respective terminals are connected to outer terminal part **901** by wiring (not illustrated) provided in FPC **90**. Inner first terminal part **902** is terminals configured to supply power to position detection element **82**.
(173) (Inner Second Terminal Part)
(174) Inner second terminal part **903** is provided in a right end portion of the front end portion of FPC **90**. In the case of the present embodiment, inner second terminal part **903** includes a plurality of terminals. These respective terminals are connected to, for example, outer terminal part **901** by wiring (not illustrated) provided in FPC **90**. Inner second terminal part **903** is terminals configured to supply power to ultrasonic motor **81**.
(175) (Left-Side Power Supply Terminal Part)
(176) Left-side power supply terminal part **92** forms a portion of a conductive path connecting inner first terminal part **902** of FPC **90** and position detection element **82**. Left-side power supply terminal part **92** is fixed to a left end portion of the rear-side surface of front-side wall part **314** (the end portion thereof on the + side in the Y direction) by a fixing means such as adhesion. Left-side power supply terminal part **92** supplies power to position detection element **82** through the left-side connection part that is not illustrated.
(177) (Right-Side Power Supply Terminal Part)
(178) Right-side power supply terminal part **93** forms a portion of a conductive path connecting inner second terminal part **903** of FPC **90** and ultrasonic motor **81**. Right-side power supply terminal part **93** is fixed to a right end portion of the rear-side surface of front-side wall part **314** (the end portion thereof on the + side in the Y direction) by a fixing means such as adhesion. Right-side power supply terminal part **93** supplies power to ultrasonic motor **81** through the right-side connection part that is not illustrated.
(179) [Lens Module]
(180) Lens module **10** is located on the + side of optical path bending module **2** in the X direction. Light from a subject enters mirror MR of optical path bending module **2** from the + side in the Z direction as indicated by long dashed short dashed line α (also referred to as the first optical axis) in FIG. **1**.
(181) The light that has entered mirror MR is bent at the optical path bending surface of mirror MR as indicated by long dashed short dashed line β (also referred to as the second optical axis) in FIG. **1** and is guided to lens part **102** of lens module **10** disposed at a stage subsequent to mirror MR (that is, on the + side thereof in the X direction).
(182) Lens module **10** includes cover **101**, a base (not illustrated), lens part **102**, and an AF device (not illustrated). Cover **101** may be integral with or separate from the cover (base main body **31** in the case of the present embodiment) of optical path bending module **2**.
(183) Lens part **102** is disposed in an accommodation space between the cover and the base in a state in which lens part **102** is held by a lens guide (not illustrated). Lens part **102** includes lens barrel **103** and one or more lenses **104** held by lens barrel **103**. Lens part **102** is supported by the base through the lens guide so as to be displaceable in the X direction.
(184) The AF device is a drive unit and displaces lens part **102** in the X direction for the purpose of auto-focusing. Note that, the structure of the AF device is not particularly limited. For example, the

AF device may be such an AF device that moves lens part **102** in the X direction with a motor (not illustrated) and by converting rotational motion of the motor into linear motion in the X direction by means of a conversion mechanism.

(185) [Image-Capturing Element Module]

(186) Image-capturing element module **11** is disposed on the + side of lens part **102** in the X direction. Image-capturing element module **11** includes, for example, an image-capturing element such as a charge-coupled device (CCD) image sensor and a complementary metal oxide semiconductor (CMOS) image sensor. The image-capturing element of image-capturing element module **11** captures a subject image formed by lens part **102** and outputs an electrical signal corresponding to the subject image. Sensor board **12** is electrically connected to image-capturing element module **11**, and power is supplied to image-capturing element module **11** through sensor board **12** and the electrical signal of the subject image captured by image-capturing element module **11** is outputted through sensor board **12**. For image-capturing element module **11** as such, a conventionally known structure can be employed.

Operation and Effect of Present Embodiment

(187) According to the present embodiment having the configuration as described above, it is possible to provide an optical actuator, a camera module, and a camera-mounted device each having a novel configuration.

Additional Notes

(188) FIGS. **12**A and **12**B illustrate automobile V as a camera-mounted device in which in-vehicle camera module vehicle camera (VC) is mounted. FIG. **12**A is a front view of automobile V, and FIG. **12**B is a rear perspective view of automobile V. In automobile V, camera module C described in the embodiment is mounted as in-vehicle camera module VC. As illustrated in FIGS. **12**A and **12**B, in-vehicle camera module VC is attached to the windshield so as to face the front side, or is attached to the rear gate so as to face the rear side, for example. This in-vehicle camera module VC is used for a rear-view monitor, a dashboard camera, collision-prevention control, automated driving control, or the like.

(189) Further, an optical actuator according to the present invention includes, as a basic configuration, an inner holder capable of holding an optical path bending member; an outer holder configured to support the inner holder in a swayable manner about a first axis; and a drive unit configured to sway the inner holder (hereinafter referred to as “the basic configuration”).

(190) Further, the optical actuator according to the present invention may have, in addition to the basic configuration described above, a configuration in which the inner holder is rotatably supported through a bearing part with respect to the outer holder at both end portions of the inner holder in a first direction parallel to the first axis (hereinafter referred to as “additional configuration 1”).

(191) Further, the optical actuator according to the present invention may have, in addition to the basic configuration described above, a configuration in which the drive unit includes an ultrasonic motor, which is supported by the outer holder, and an interposition part, which is supported by the inner holder, the ultrasonic motor includes a transducer configured to resonate, and the interposition part includes a contact part configured to come into contact with the transducer and having a fan shape (hereinafter referred to as “additional configuration 2”).

(192) Further, the optical actuator according to the present invention may have, in addition to the basic configuration described above, a configuration in which the bearing part includes: an inner ring including an inner ring raceway in an outer peripheral surface of the inner ring; an outer ring including an outer ring raceway in an inner peripheral surface of the outer ring; and a plurality of rolling elements rollably provided between the inner ring and the outer ring (hereinafter referred to as “additional configuration 3”).

(193) Note that, the present invention may include, in addition to the basic configuration described above, at least one configuration of additional configurations 1 to 3 described above. Further, the

present invention may include, in addition to the basic configuration described above, a configuration in which additional configuration 1 to 3 are appropriately combined.
(194) The disclosure of U.S. Provisional Application No. 63/119,001, filed on Nov. 30, 2020, including the specification, drawings and abstract, is incorporated herein by reference in its entirety.
INDUSTRIAL APPLICABILITY
(195) An optical actuator and a camera module according to the present invention can be mounted in, for example, a thin camera-mounted device such as a smartphone, a mobile phone, a digital camera, a notebook computer, a tablet terminal, a portable game machine, and an in-vehicle camera, for example.
REFERENCE SIGNS LIST
(196) C Camera module **2** Optical path bending module **3** Base **31** Base main body **311** Lower-side wall part **312***a* Left-side wall part **313***b* Right-side wall part **314** Front-side wall part **315** Upper-side wall part **316** Upper-side opening part **317** Front-side opening part **318** Rear-side opening part **321** Lower-side opening part **36** Rear-side plate MR Mirror S Mirror swaying device **4** Inner holder **41** Mount part **41***a* Left-side plate part **42***b* Right-side plate part **43** Rear-side plate part **44***a* Left-side shaft part **45***b* Right-side shaft part **46** Inner fixing part **47** Magnet disposing part **5** Outer holder **5***c* Accommodation space **501** Inclined plate part **502***a* Left-side plate part **503***b* Right-side plate part **504** Front-side plate part **505** Rear-side plate part **506** Lower-side plate part **507** Front-side opening part **508***a* Left-side bearing holding part **509***b* Right-side bearing holding part **514** Fixing part **515** Element disposing part **63** Sway support part **631***a* Left-side bearing **632***b* Right-side bearing **8** Drive unit **80** Interposition part **801** Holder **802**, **803** Contact part **802***a*, **803***a* Arc part **81** Ultrasonic motor **810** Resonance part **811** Trunk part **812**, **813** Transducer **814** Energization part **815**, **816** Piezoelectric element **817** First electrode **818** Sandwiching part **819** Electrode part **82** Position detection element **83** Magnet **9** Power supply part **90** FPC **901** Outer terminal part **902** Inner first terminal part **903** Inner second terminal part **92** Left-side power supply terminal part **93** Right-side power supply terminal part **10** Lens module **101** Cover **102** Lens part **103** Lens barrel **104** Lens **11** Image-capturing element module **12** Sensor board **13** Control unit A**1** First axis V Automobile VC In-vehicle camera module M Smartphone

Claims

1. An optical actuator, comprising: an inner holder comprising a plated-shaped mount part including an upper surface and a lower surface and capable of holding an optical path bending member on the upper surface, the optical path bending member being configured to make a light entering in a Z direction emit in an X direction; an outer holder comprising: a pair of side plate parts configured to support the inner holder in a swayable manner about a first axis extending in parallel to a Y direction; and an inclined plate part provided between the pair of side plate parts in the Y direction and forming an accommodation space for the inner holder together with the pair of side plate parts; and a drive unit configured to sway the inner holder, wherein: the inner holder is rotatably supported through a bearing part with respect to the pair of side plate parts of the outer holder at both end portions of the inner holder in the Y direction, the drive unit includes an ultrasonic motor and an interposition part, the ultrasonic motor being supported by the outer holder, the interposition part being supported by the inner holder, the upper surface and the lower surface of the plated-shaped mount part are disposed to extend to be in parallel to the Y direction and inclined with respect to the X direction and the Z direction, the inclined plate part includes an upper surface disposed to extend to be in parallel to the Y direction and inclined with respect to the X direction and the Z direction to face the lower surface of the plated-shaped mount part to form an inclined gap therebetween, the ultrasonic motor includes a resonator including a transducer configured to resonate, the resonator being a plate-shaped member disposed on the upper surface of

the inclined plate part in the inclined gap, and the interposition part includes a contact part disposed the inclined gap and configured to come into contact with the transducer, the contact part having a fan shape.

2. The optical actuator according to claim 1, wherein: the ultrasonic motor includes a pair of the transducers, and the interposition part includes a pair of the contact parts configured to respectively come into contact with the pair of transducers.

3. The optical actuator according to claim 1, wherein the contact part includes an arc part formed of an arc centered on a center point on the first axis.

4. The optical actuator according to claim 3, wherein: the transducer is disposed so as to be along a tangent to a circle around the first axis, and when the transducer resonates, a driving force in a direction parallel to the tangent acts on the contact part from the transducer.

5. The optical actuator according to claim 1, wherein: the bearing part includes: an inner ring including an inner ring raceway in an outer peripheral surface of the inner ring; an outer ring including an outer ring raceway in an inner peripheral surface of the outer ring; and a plurality of rolling elements rollably provided between the inner ring and the outer ring.

6. A camera module, comprising: the optical actuator according to claim 1; the optical path bending member held by the inner holder of the optical actuator; and an image-capturing element disposed on a front side with respect to the optical actuator in the X direction.

7. A camera-equipped device, comprising: the camera module according to claim 6; and a control unit configured to control the camera module.